



ECE297 Team 033

EssenceNav

Andrew, Xin



Content Table

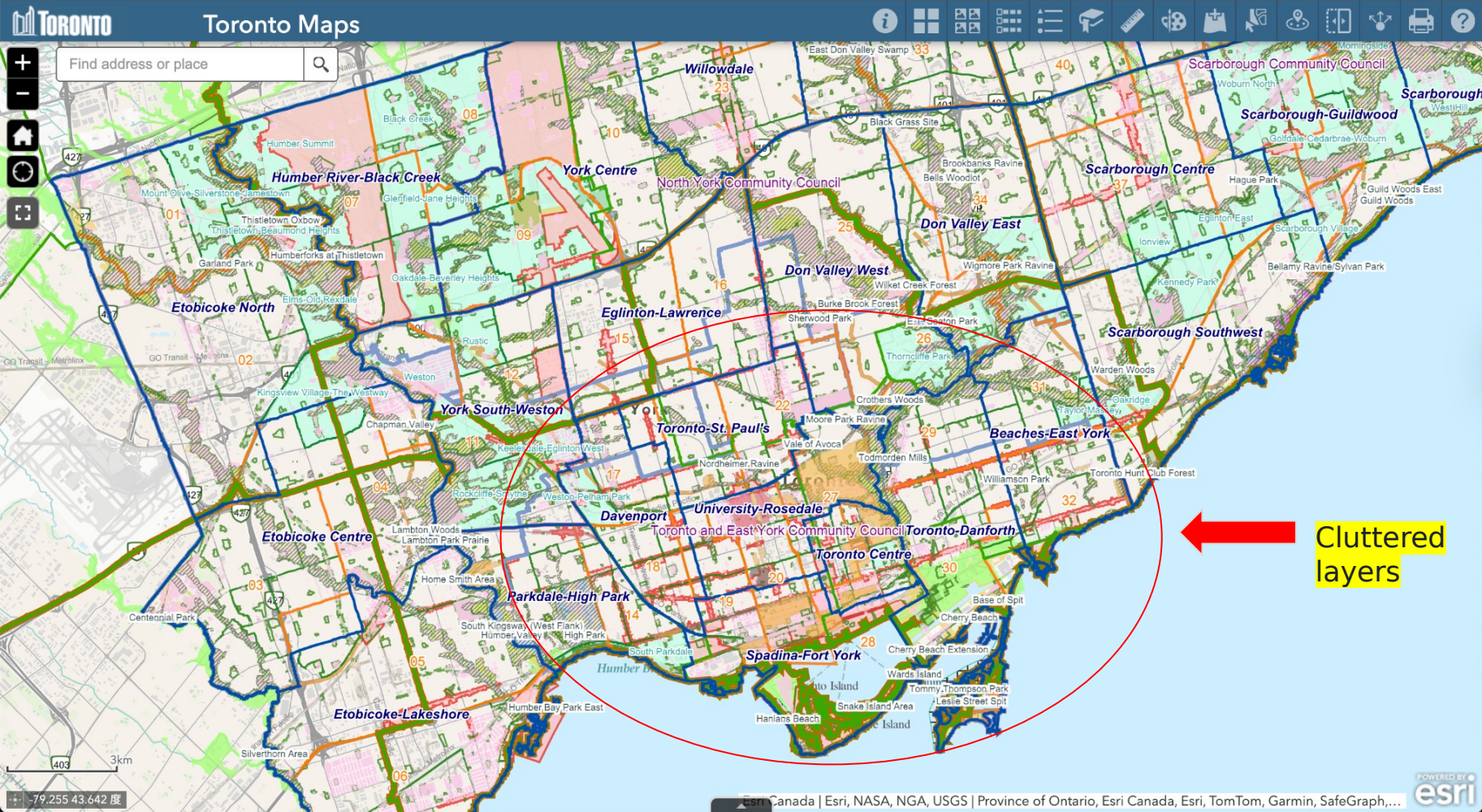
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- II. Explanation and Demonstration
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 - B. Innovative Features
 - 1. 2.5D Visual
 - 2. AI assistant
 - 3. Multi-Language
 - C. Algorithms
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- V. Reference



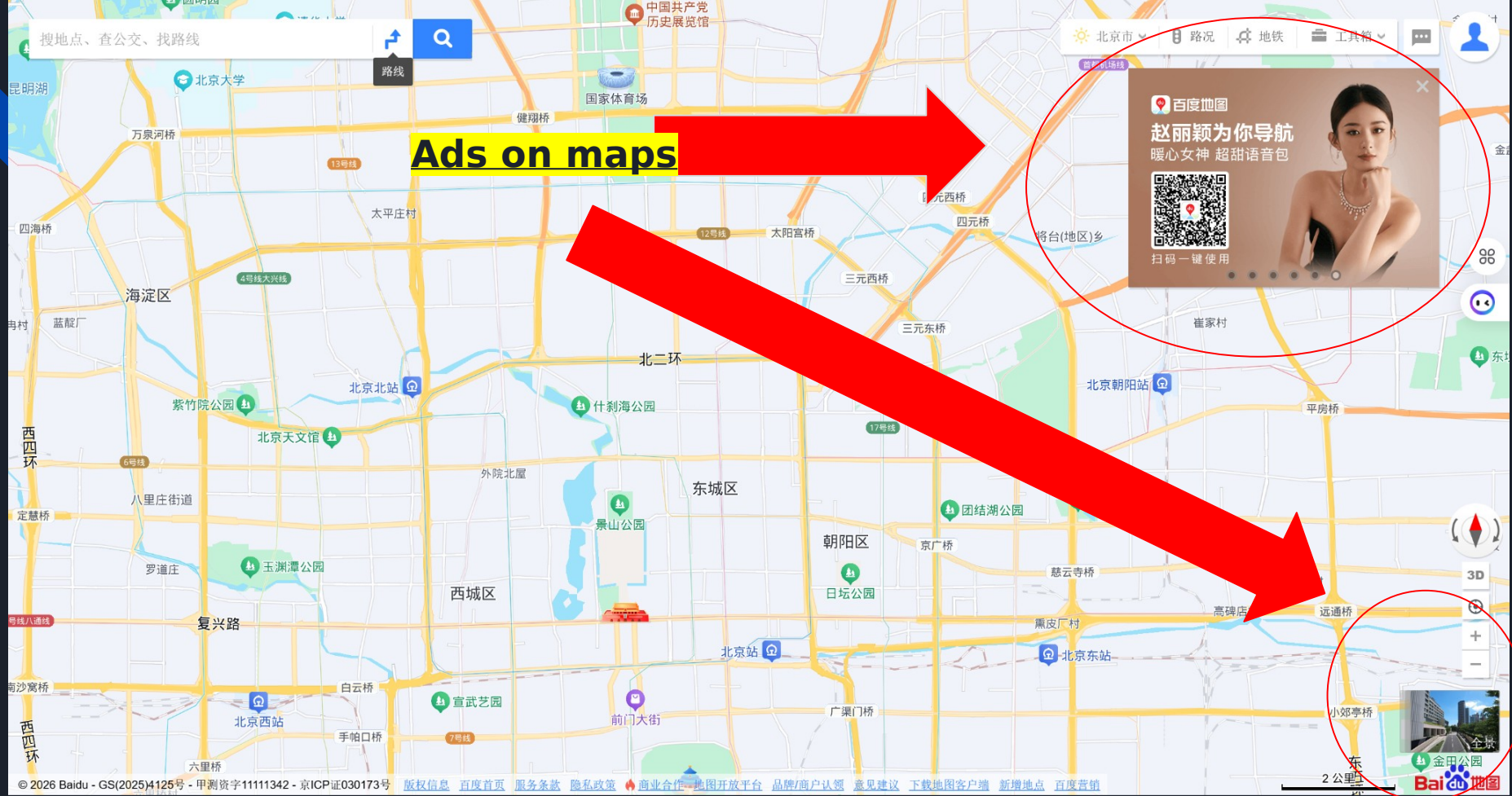
I. GAP



More information does not always mean
better navigation



Cluttered
layers





Information overload has a user cost

Tourists

- Unfamiliar with the city
- Need quick orientation, not dense map layers

Busy commuters

- Already under time pressure
- Need fast decisions with minimal distraction



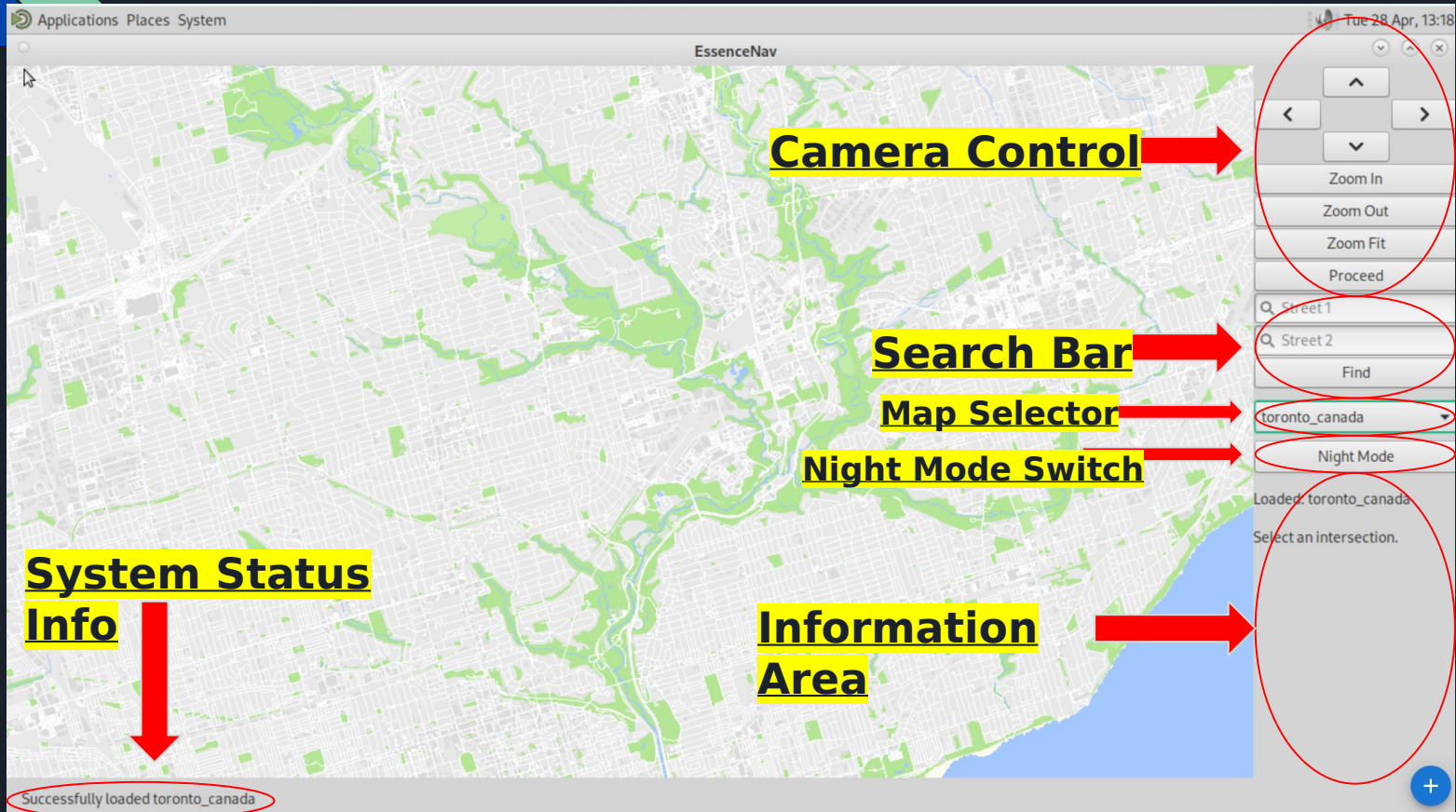
Simplicity

Addressing Information
Overload



II. Explanation & Demonstration

Map Overview (Basic Features)



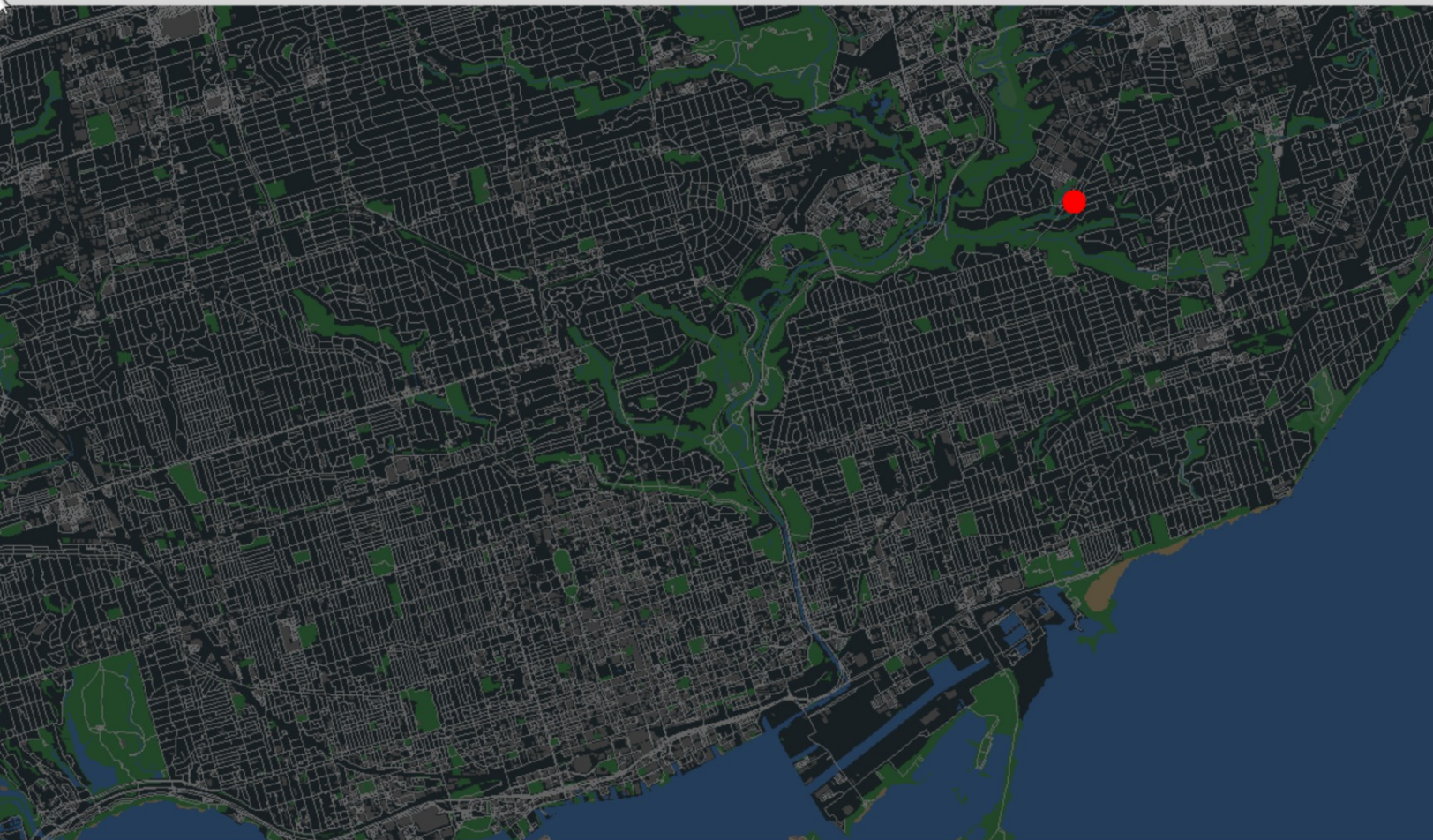


Basic Features

1. Night Mode
2. Auto Zoom
3. Intuitive Camera Control and LOD
4. Search Bar & Autocomplete
5. “Loading” Information
6. Path Finding with Different Modes, Navigational Instruction
(M3)



Night Mode



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Zoom In

Zoom Out

Zoom Fit

Proceed

Q Street 1

Q Street 2

Find

toronto_canada

Night Mode

It looks like one of the streets at this intersection is missing a name in the map data—there's an unnamed road crossing St Clair Avenue East near that area. This intersection has three connecting road segments, but without a name for the other street, it may be hard to find exactly. You might try searching for a nearby street name or landmark instead.

Different Color when Different Modes

Night Mode

Yes!

NO

```
switch (feat.type)
{
case PARK:
g->set_color(g_night_mode ? ezgl::color(34, 70, 40) : ezgl::color(180, 230, 150)); // green
break;
case GREENSPACE:
g->set_color(g_night_mode ? ezgl::color(34, 70, 40) : ezgl::color(180, 230, 150)); // green
break;
case GOLFCOURSE:
g->set_color(g_night_mode ? ezgl::color(46, 76, 51) : ezgl::color(194, 235, 167)); // green
break;
case LAKE:
g->set_color(g_night_mode ? ezgl::color(36, 61, 91) : ezgl::color(163, 200, 254)); // blue
break;
case RIVER:
g->set_color(g_night_mode ? ezgl::color(36, 61, 91) : ezgl::color(163, 200, 254)); // blue
break;
case STREAM:
g->set_color(g_night_mode ? ezgl::color(36, 61, 91) : ezgl::color(163, 200, 254)); // blue
break;
case BEACH:
g->set_color(g_night_mode ? ezgl::color(90, 80, 60) : ezgl::color(255, 250, 240)); // light sand
break;
case ISLAND:
```



Camera Controls and Level of Details

How Implemented

Our solution

Level of Detail [4] for
:

What might happen?

Icons touching and overlapping each other



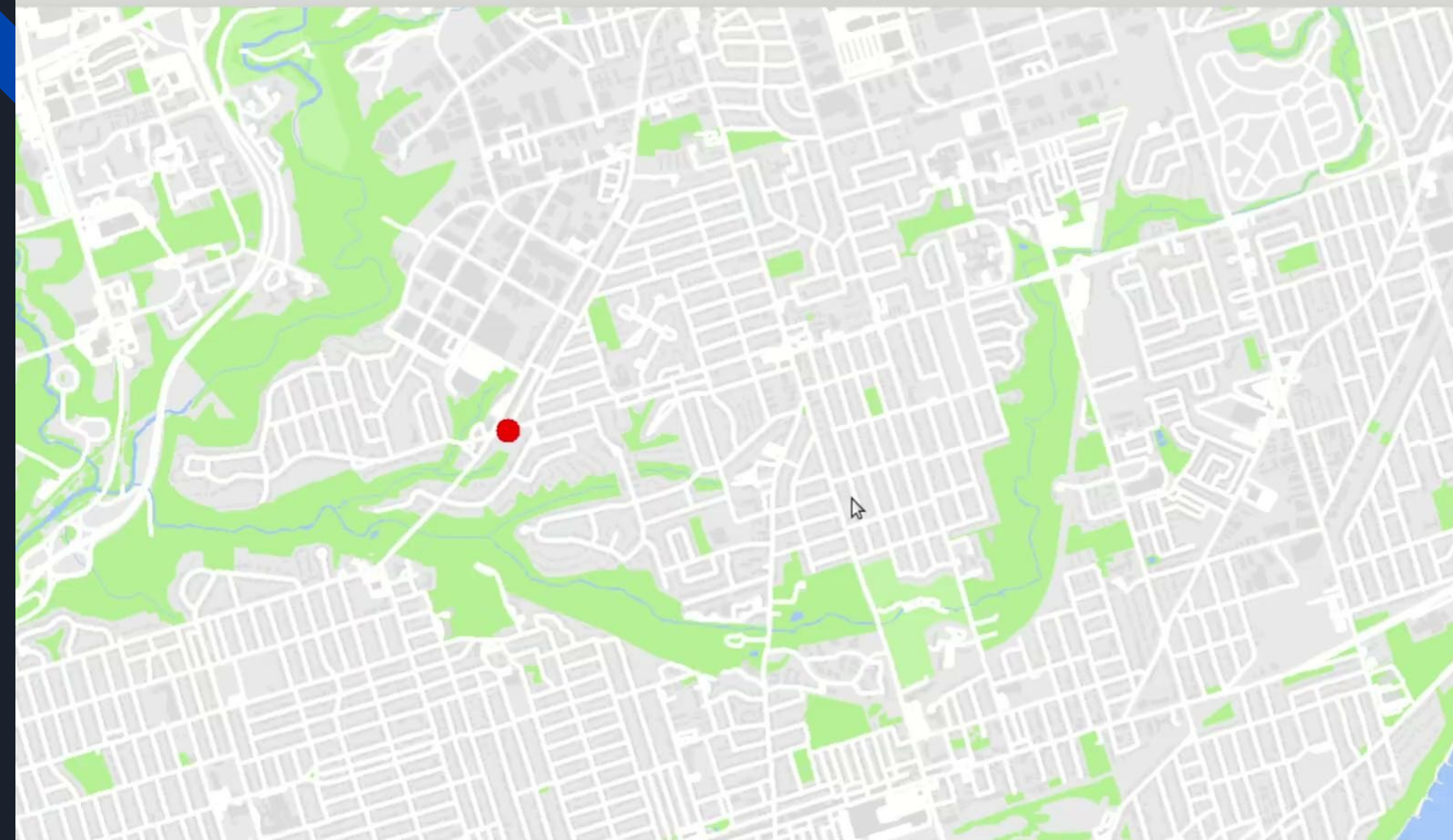
Why important?

Users cannot effectively processing the information



- 2.5D buildings
- POI
- POI name
- One-way indicator
- Street name

EssenceNav



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Zoom In

Zoom Out

Zoom Fit

Proceed

Q Street 1

Q |

Find

toronto_canada

Night Mode

It looks like one of the streets at this intersection is missing a name in the map data—there's an unnamed road crossing St Clair Avenue East near that area. This intersection has three connecting road segments, but without a name for the other street, it may be hard to find exactly. You might try searching for a nearby street name or landmark instead.

Successfully loaded toronto_canada





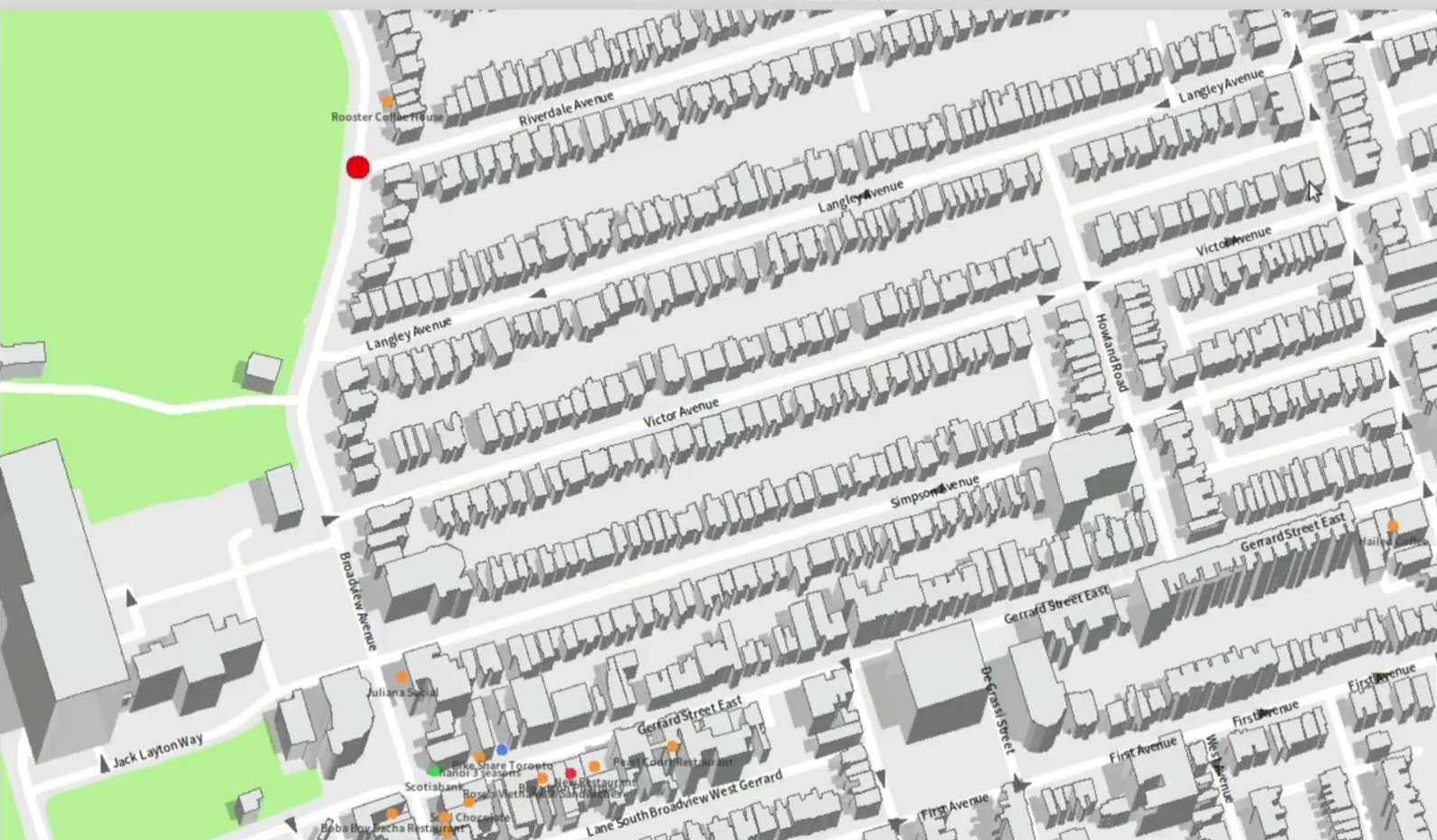
Search Bar & Auto Complete



Why Implemented: Reducing search friction and input errors with Autocomplete in Search Bar

- Problem: Users may not know the exact street name or spelling.
- Design: Autocomplete suggests matching street names as the user types.

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Navigation controls:

- Zoom In
- Zoom Out
- Zoom Fit
- Proceed

Search bar:

Q Victor A

Q Howland R

Find

toronto_canada

Night Mode

You're looking at the intersection of Broadview Avenue and Riverdale Avenue in Toronto. Three street segments meet here.

Successfully loaded toronto_canada

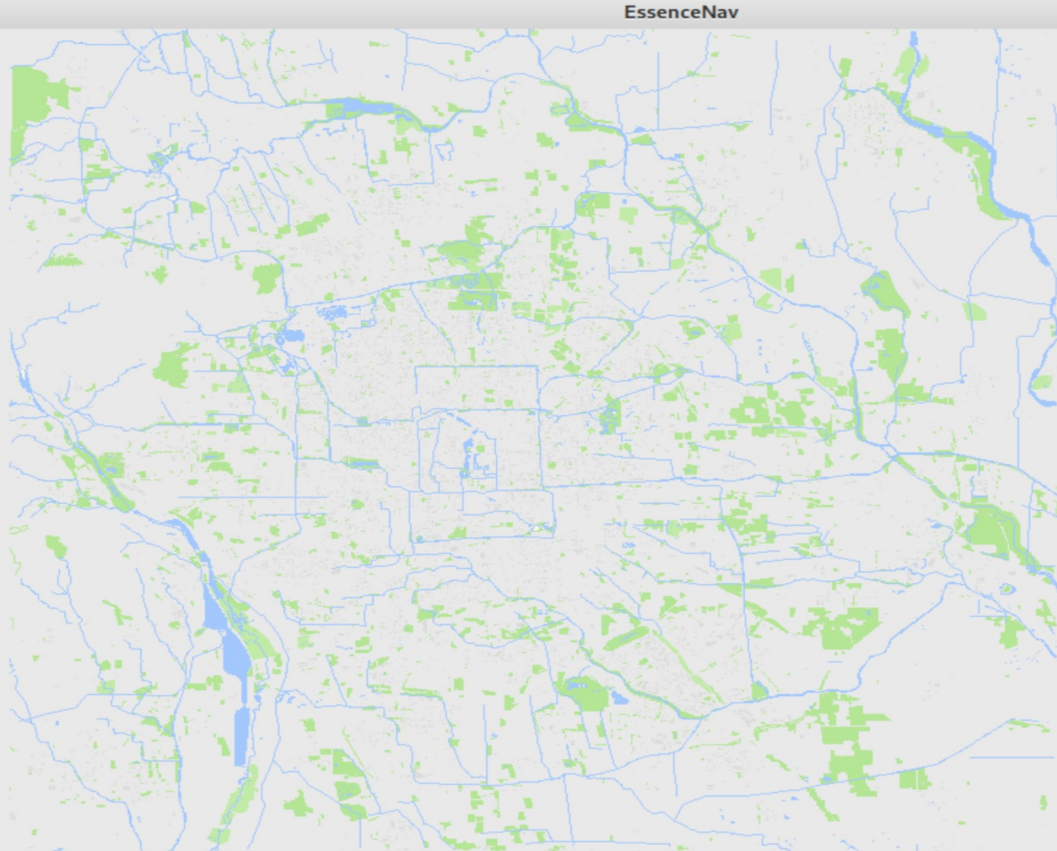


Loading Hints

**Hint for
Loading**

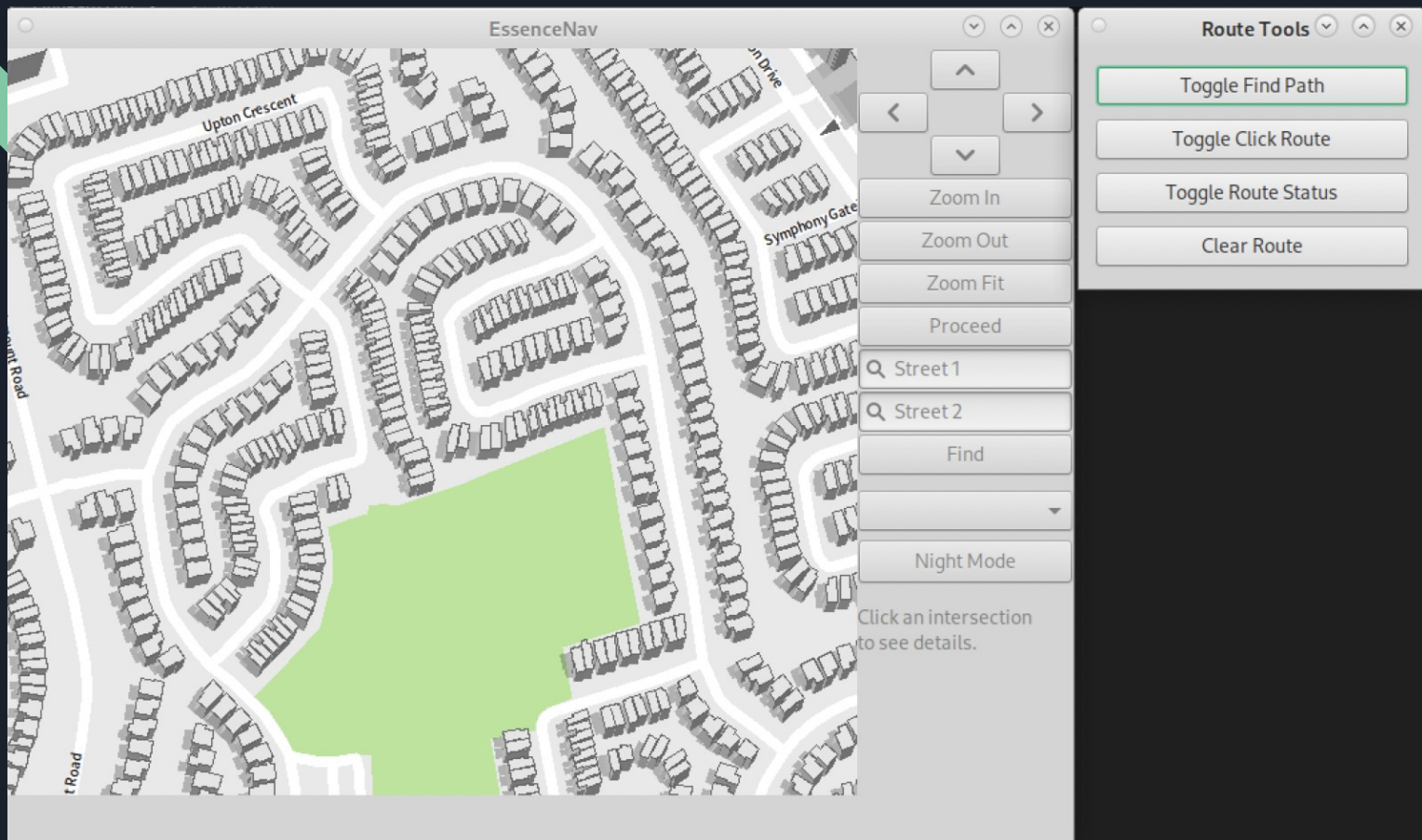


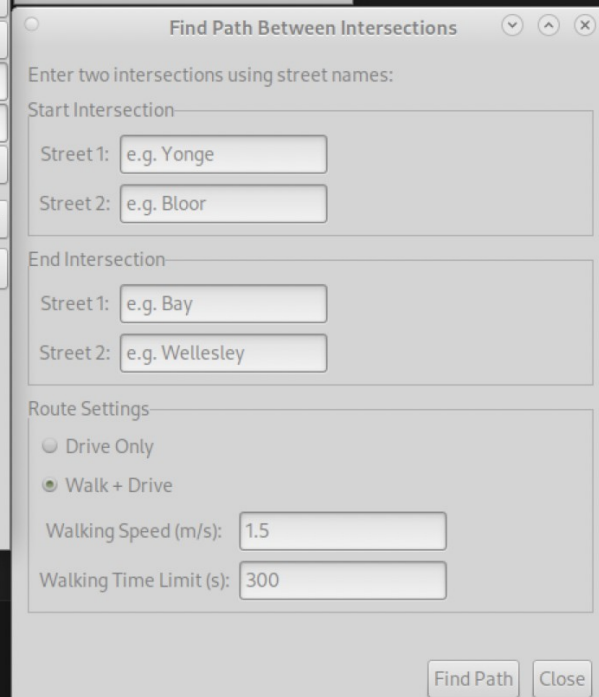
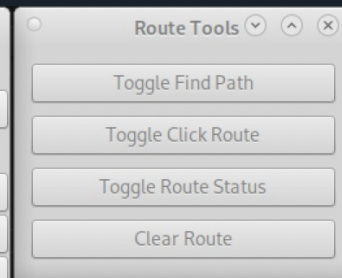
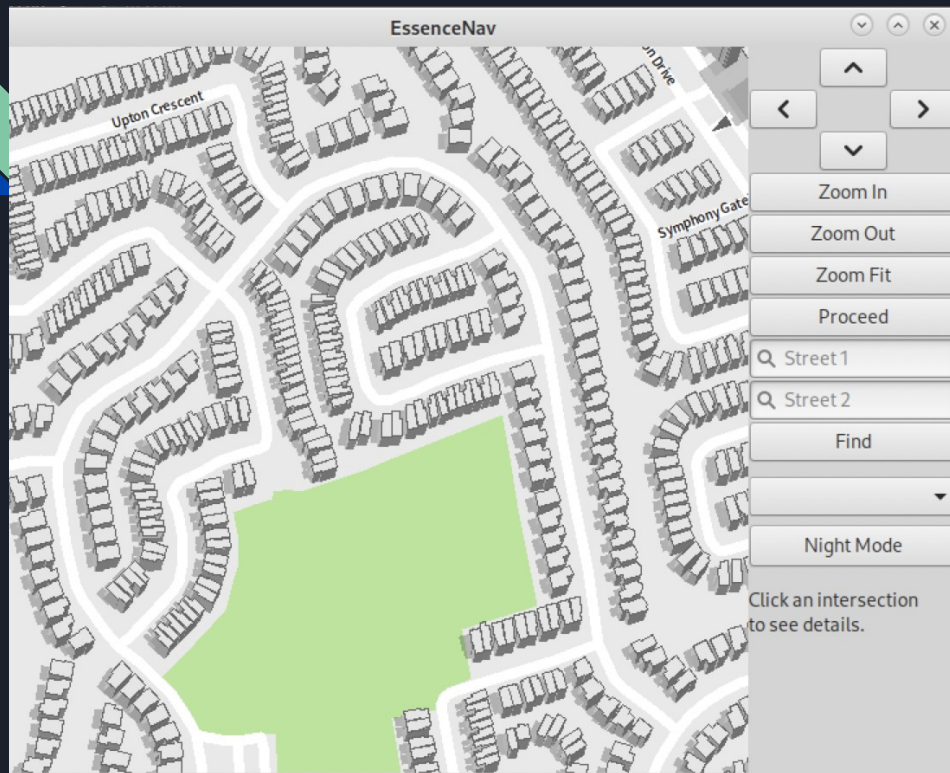
Loading tokyo_japan...





M3 Implementation





```
ty())  
0.0;
```

CONSOLE TERMINAL PORTS

```
s/public/maps/toronto_canada.streets.bin  
for 1192954 ways
```

EssenceNav



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Zoom In

Zoom Out

Zoom Fit

Proceed

🔍 Street 1

🔍 Street 2

Find

▼

Night Mode

Click an intersection
to see details.

Route Tools ⌵ ⌶ ⌵

Toggle Find Path

Toggle Click Route

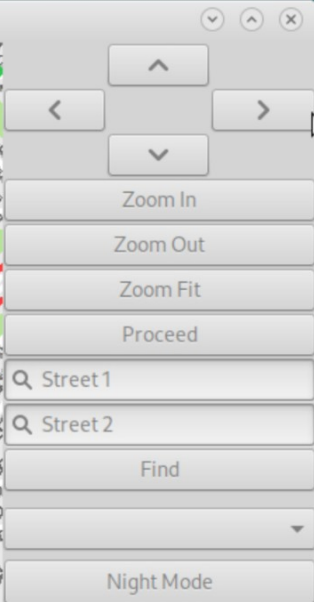
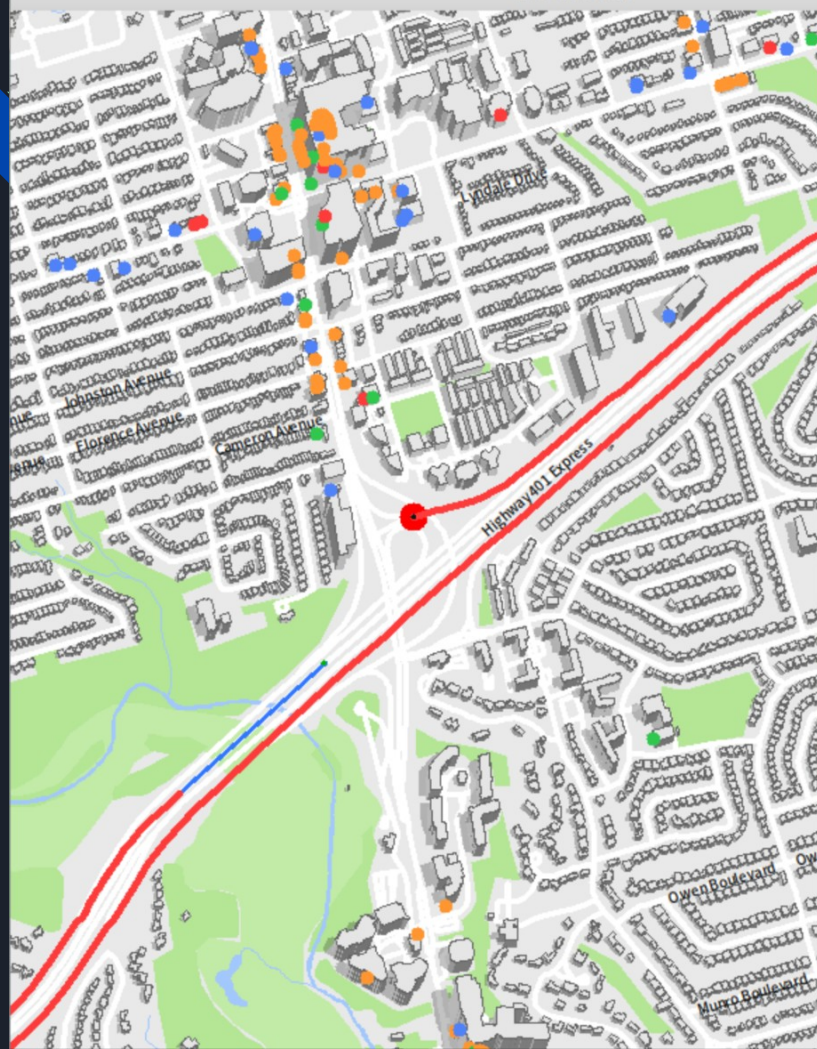
Toggle Route Status

Clear Route

Click Route Selection ⌵ ⌶ ⌵

Press 'Start Click Selection', then click a start and destination on the map.
Press 'Start Click Selection', then click a start and destination on the map.

Start Click Selection



found a spot at coordinates (43.7553, -79.4078) with an unknown name, located where two street segments meet. It only intersects with one other street, also unnamed—would you like to search for a nearby address or landmark instead?

Mode: Walk + Drive
Walking time: 4 min 45 s
Driving time: 8 min 57 s

Walk to pick-up point first.

Travel Directions:

Toggle Find Path

Toggle Click Route

Toggle Route Status

Clear Route

Click Route Selection

Press 'Start Click Selection', then click a start and destination on the map.

Path found successfully.

Start Click Selection

Confirm Route

Route Status

Mode: Walk + Drive

Drive segments: 39

Walk segments: 1

Clear Route



Innovative Features

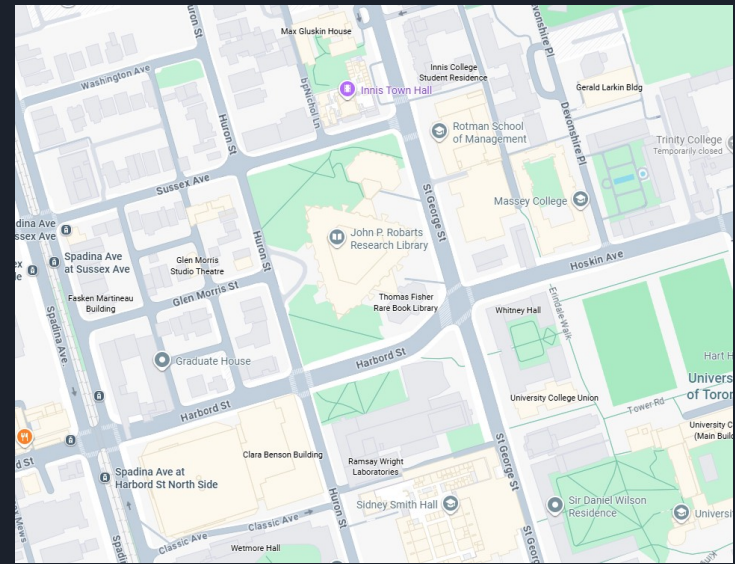
1. 2.5 D visual
2. AI assistant
3. Multi-language support



2.5D Visual

Why Implemented

- 2D is Top-Down view
- Depth and direction
- Spatial Orientation
- Reduce cognitive load



[5]M.-C. Zheng and Y.-W. Hsu, "How 2.5D Maps Design Improve the Wayfinding Performance and Spatial Ability of Map Users," *Informatics*, vol. 8, no. 4, p. 88, Dec. 2021, doi: <https://doi.org/10.3390/informatics8040088>.



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Zoom In

Zoom Out

Zoom Fit

Proceed

Find

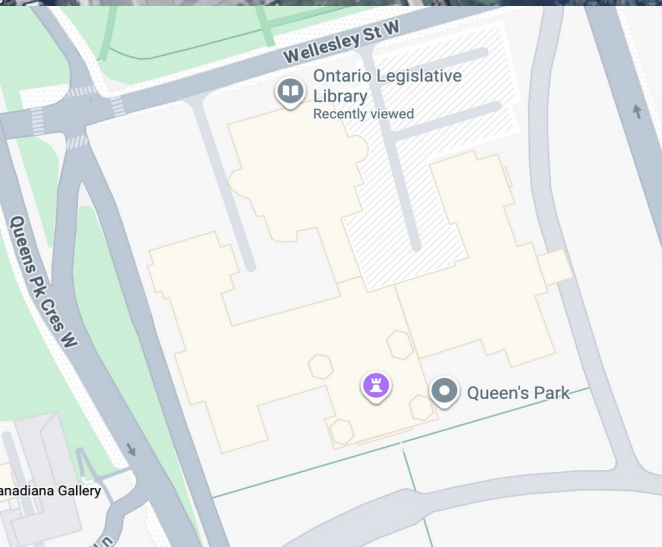
Night Mode

I found a spot at coordinates (43.7553, -79.4078) with an unknown name, located where two street segments meet. It only intersects with one other street, also unnamed—would you like to search for a nearby address or landmark instead?

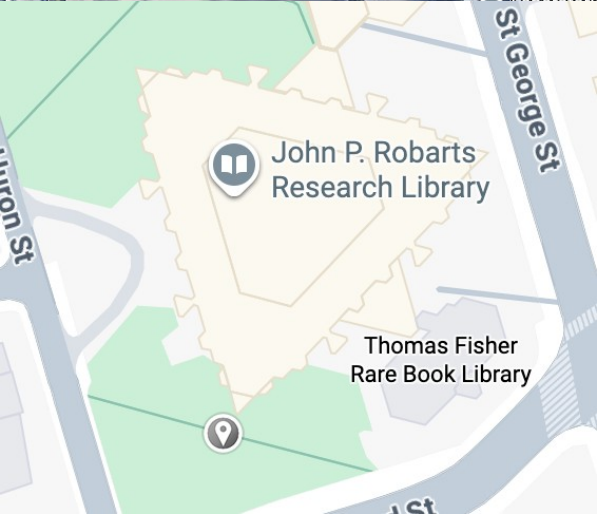
Walk to pick-up point first.

Travel Directions:

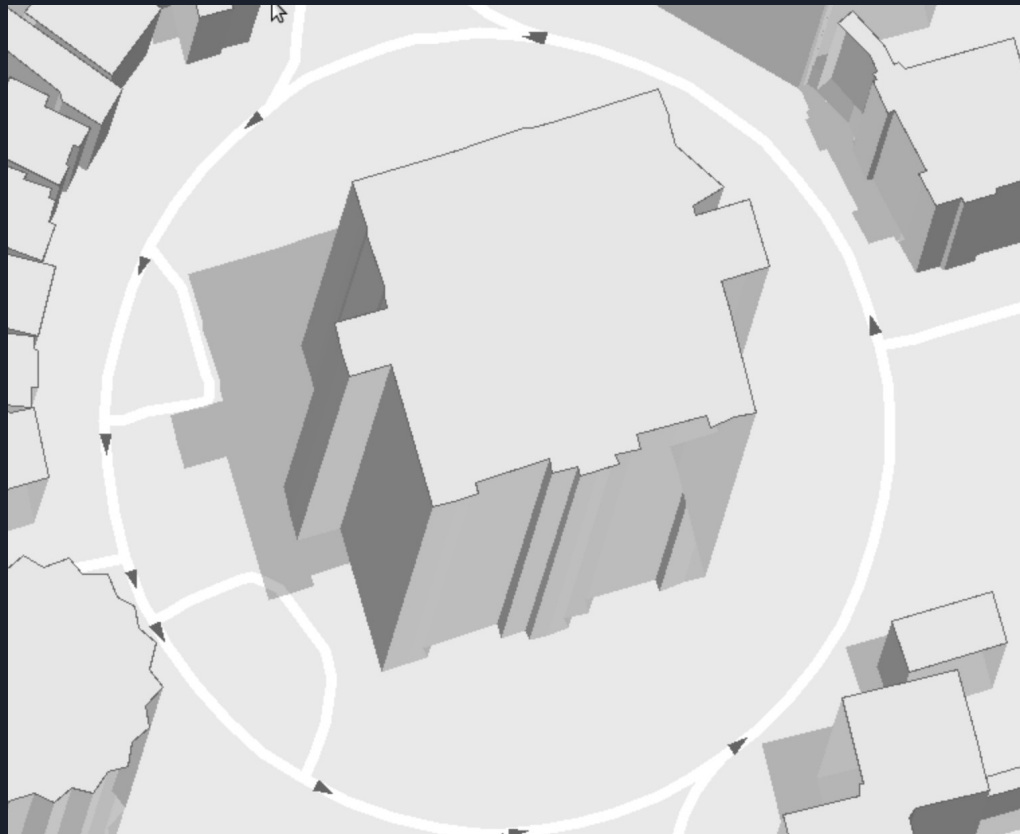
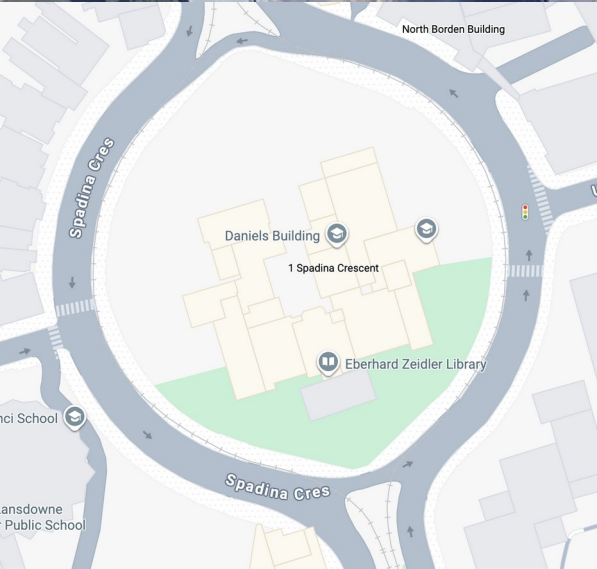
1. Go straight on Highway 401 Express for 60.8 meters.



Ontario Legislative Assembly [3]



Robarts Research Library [3]



Faculty of Architecture [3]

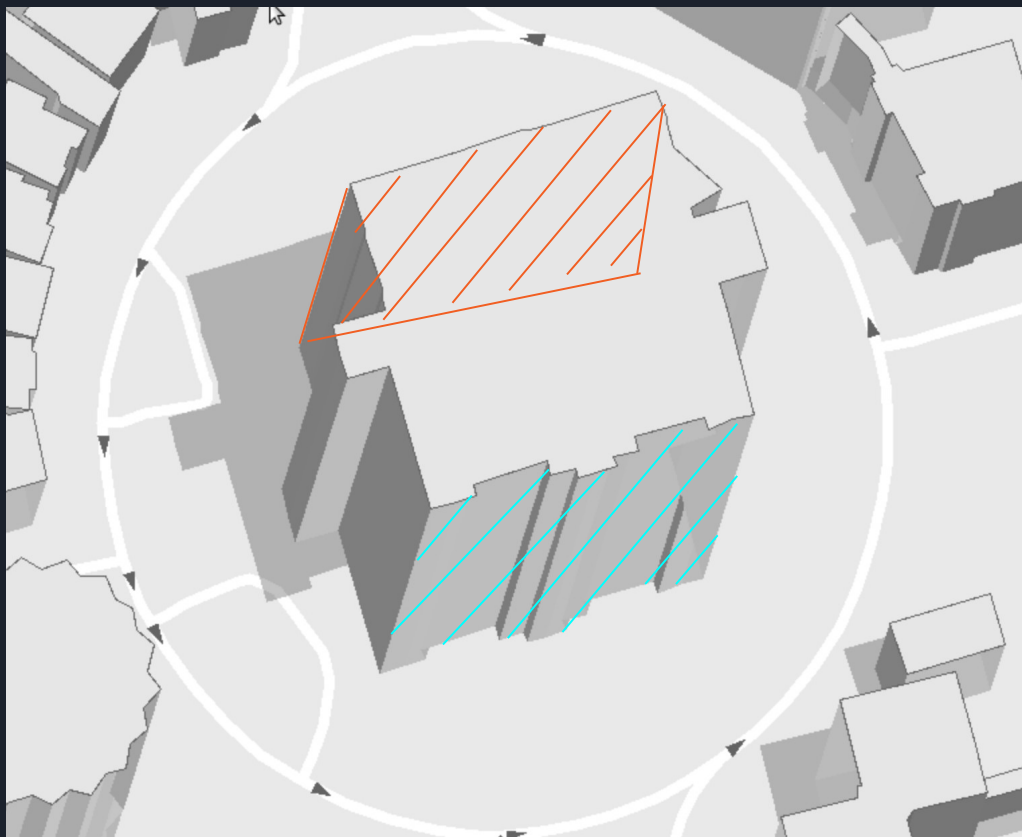


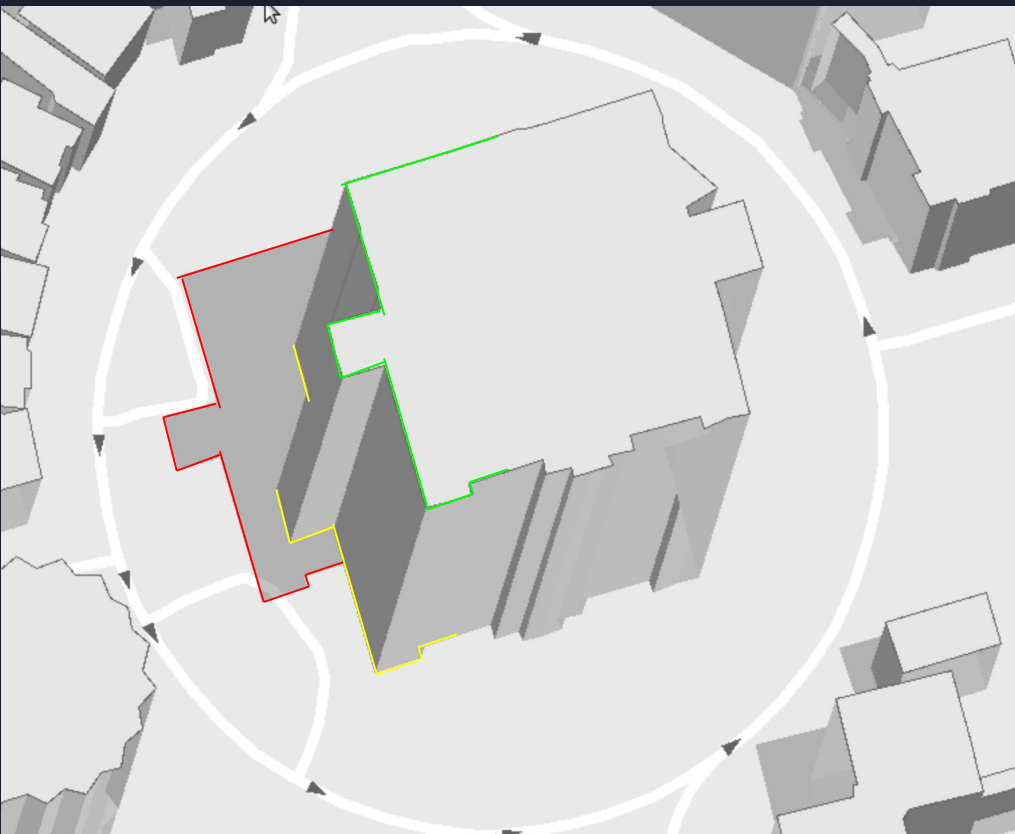
How Implemented

Depth Sorting

Drop Shadow

Directional Light





Drop Shadow

- Original Polygon
- Drop Shadow
- Roof

```

for (size_t i = 0; i < feat->points.size() - 1; ++i)
{
    double dx = feat->points[i + 1].x - feat->points[i].x;
    double dy = feat->points[i + 1].y - feat->points[i].y;
    double len = std::hypot(dx, dy);
    if (len == 0)
        continue;

    // Outward normal vector
    double nx = dy / len;
    double ny = -dx / len;

    // Simple directional light from North-West
    double light_x = 0.8;
    double light_y = -0.6;

    // dot product with light
    double dot = nx * light_x + ny * light_y;
    double intensity = 0.85 + 0.25 * dot;

    int r = std::min(255, std::max(0, int(base_wall.red * intensity)));
    int g_col = std::min(255, std::max(0, int(base_wall.green * intensity)));
    int b_col = std::min(255, std::max(0, int(base_wall.blue * intensity)));

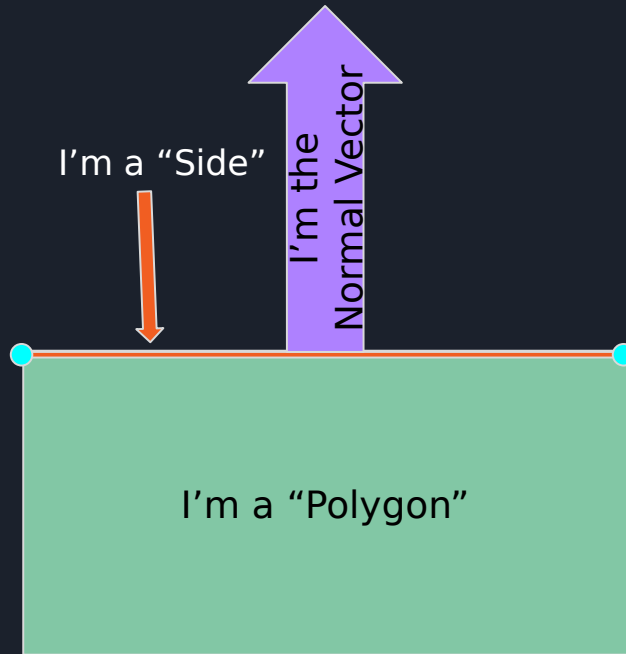
    g->set_color(r, g_col, b_col);

    std::vector<ezgl::point2d> wall = {
        feat->points[i], feat->points[i + 1], roof_pts[i + 1], roof_pts[i]};
    g->fill_poly(wall);
}

```

Directional Light

1. Calculate the normal vector of the wall
2. Directional light vector
3. Dot product for intensity



Sun Vector

Vector B

**Wall Normal
Vector**

Vector A

$$\text{Dot Product} = |A||B|\cos(\text{Angle between Them}) = A.x * B.x + A.y * B.y$$



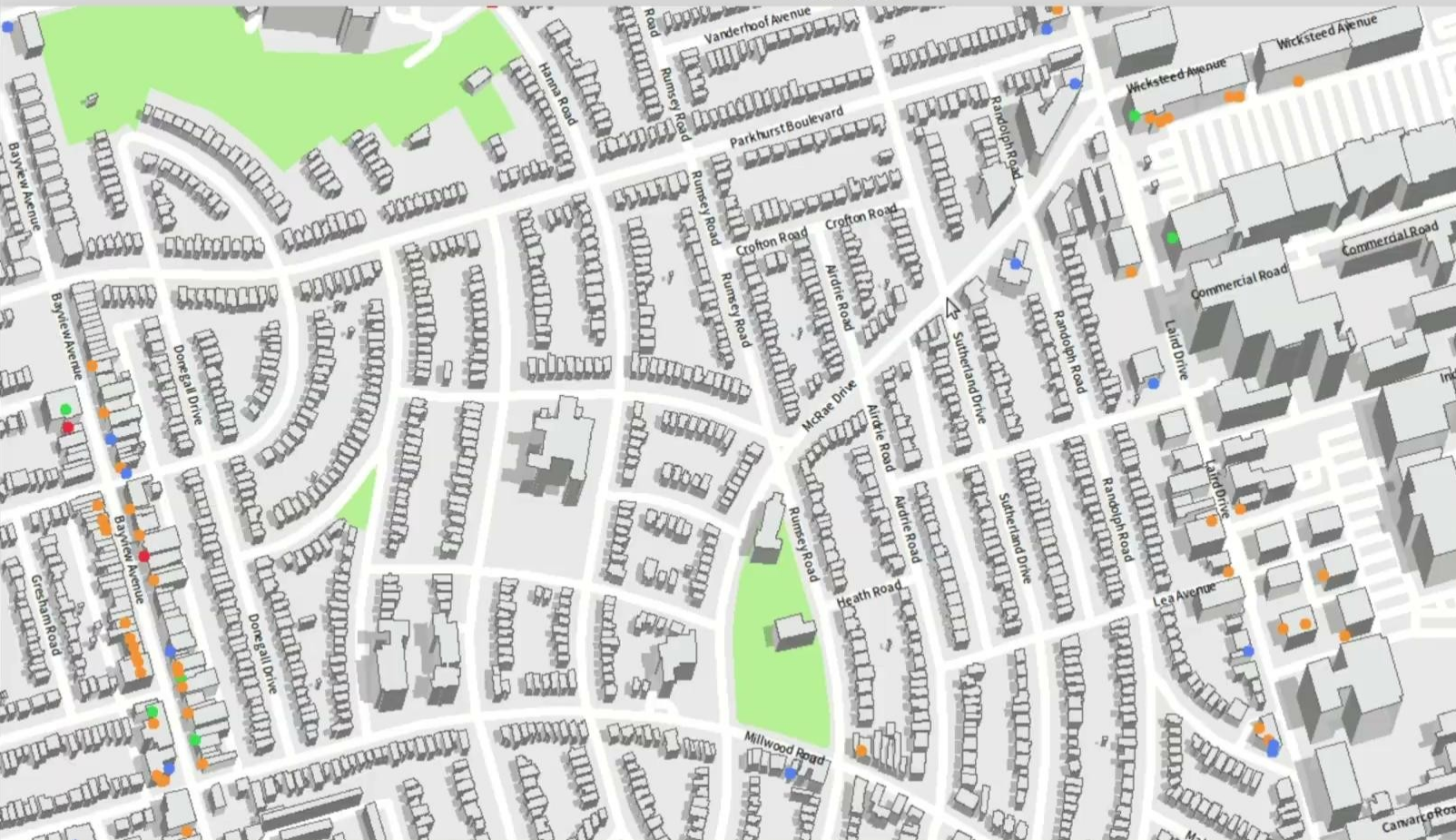
Gen-AI Assistant



Reducing interpretation effort with AI Assistant

- Raw map feedback can be easy to miss.
- Summarizing selected roads and intersections in natural language.
- Users understand without reading debug-style output.
- Optional layer: core map still works without it

EssenceNav



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Zoom In

Zoom Out

Zoom Fit

Proceed

Q Street 1

Q Street 2

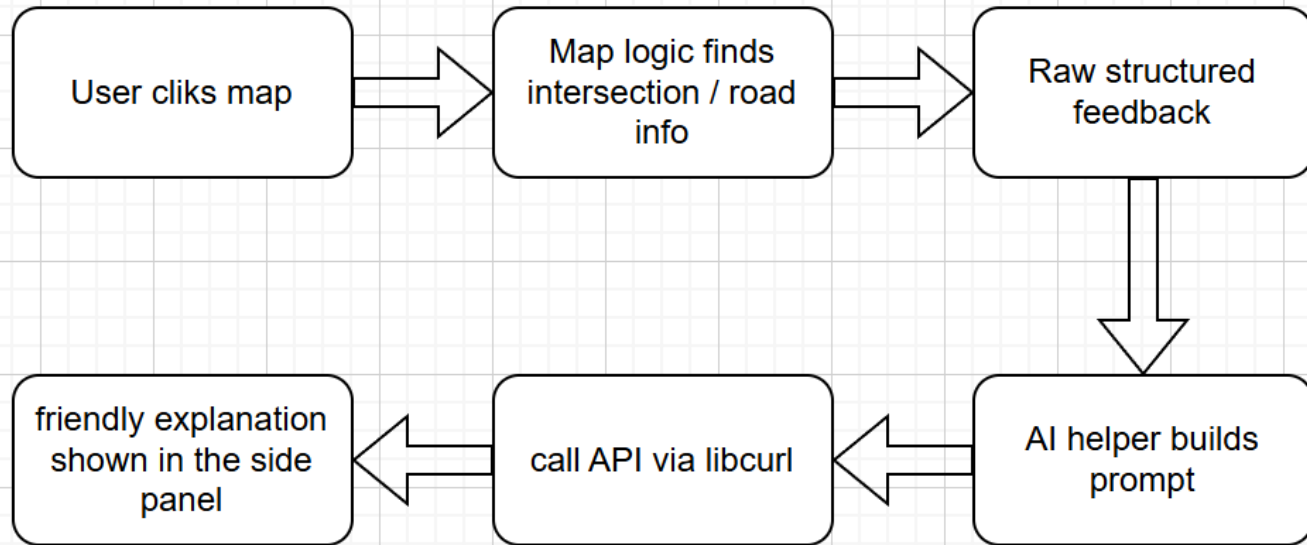
Find

Night Mode

There's an intersection at Hanna Road and Parklea Drive. It connects to four road segments nearby.



Assistant pipeline: from map data to readable feedback





Multi-language Support

Non-Latin scripts supported

Roads displayed correctly in Mandarin

Successfully loaded beijing_china

Non-Latin scripts supported

Roads displayed correctly in Mandarin

Successfully loaded beijing_china

Non-Latin scripts supported

Roads displayed correctly in Mandarin

Successfully loaded beijing_china

Latin Scripts supported

Correct display in Spanish

Latin Scripts supported

EssenceNav

Correct display in Spanish

rio-de-janeiro_brazil

Find

Night Mode

Loaded: rio-de-janeiro_brazil

Select an intersection.



Why it matters

- For locals:

Matches real-world street signs.

- For tourists:

Local language keeps consistency

English fallback keeps readability.



Language Display Strategy

1. Prefer local names: consistency with real-world signs.
2. If unavailable, fallback to English
3. Preserve original data: Reduces ambiguity across regions.



Algorithms



Algorithms

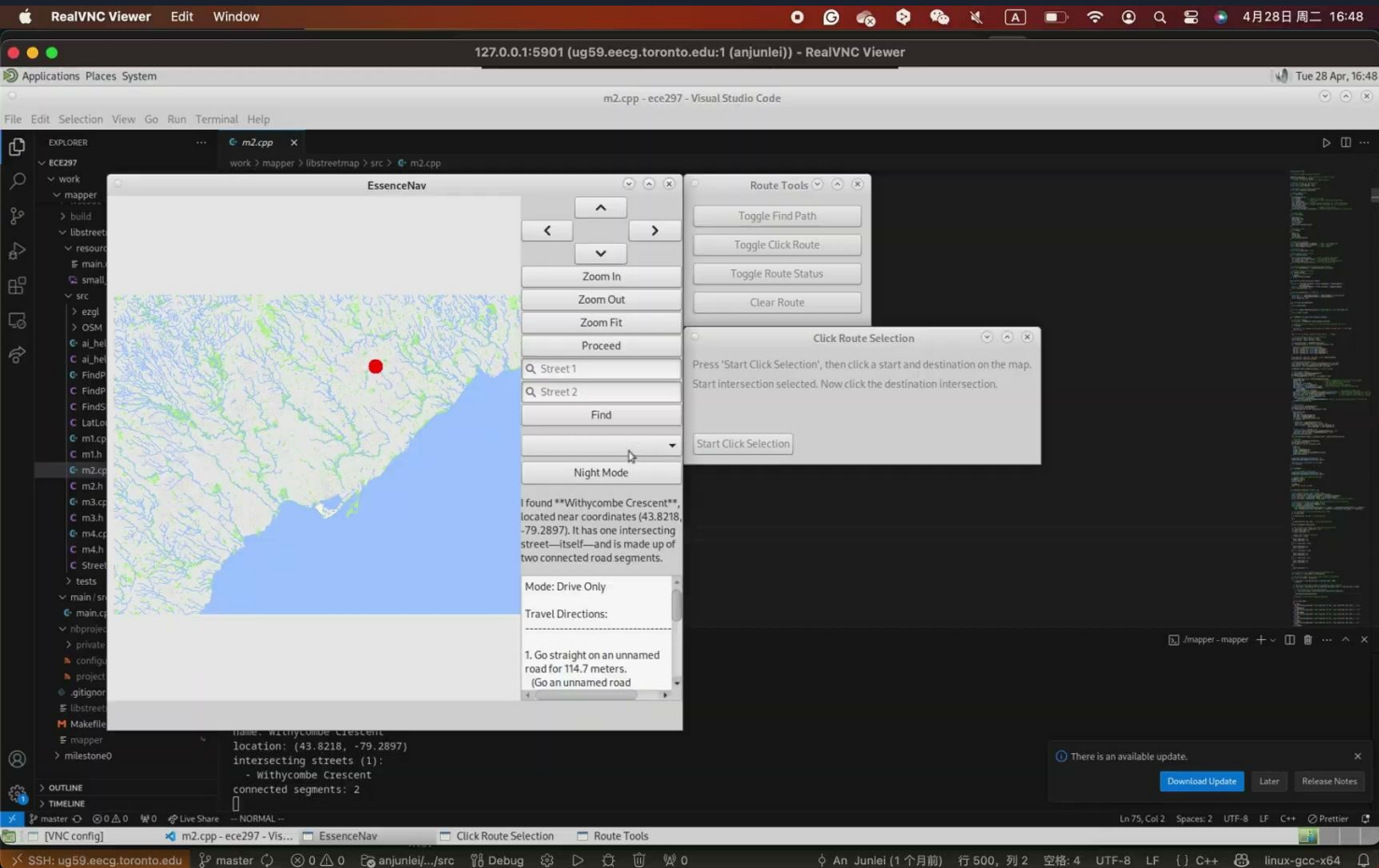
M3: Dijkstra's algorithm

M4: Greedy Algorithm + 2-opt + parallel computation +
cached results

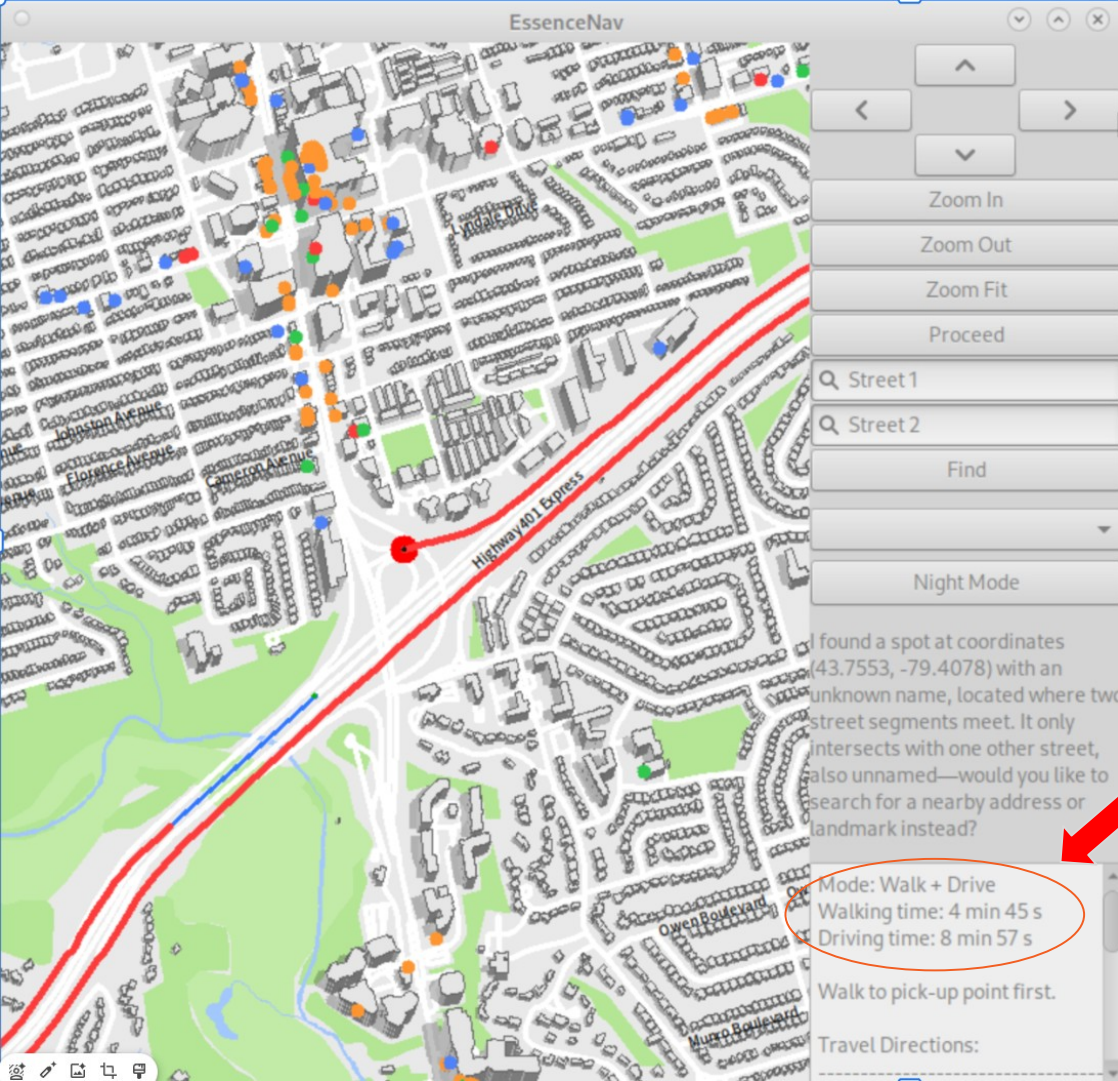


Milestone 3

Our pathfinding algorithm
meets...



Low time-complexity, Optimal run time



Real-life based Time Estimation



Practical Uses

- | | |
|---|--|
| 1. Go straight on Withycombe Crescent for 300.3 meters. | 4. Turn Right onto Artillery Street. |
| 2. Turn Left onto Port Royal Trail. | 5. Go straight on Artillery Street for 230.4 meters. |
| 3. Go straight on Port Royal Trail for 249.3 meters. | 6. Turn Left onto Midland Avenue. |
| | 7. Go straight on Midland Avenue for 1710.4 meters. |

Matches a driver's need for practical, easy-to-follow directions.



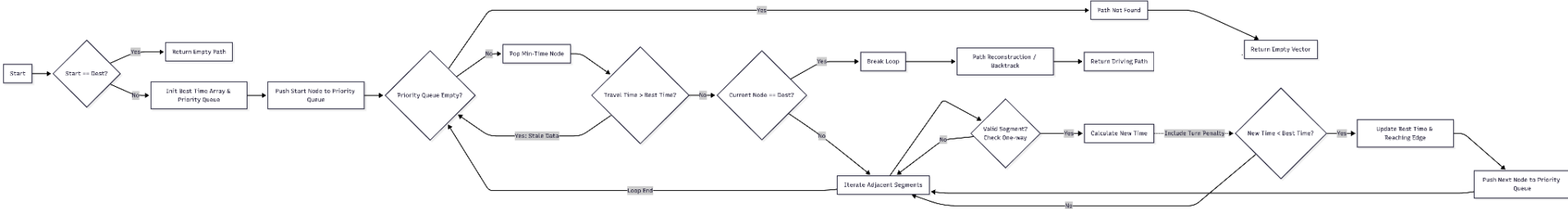
Dij (Priority Queue with Min Heap, and OPTIMIZATION)

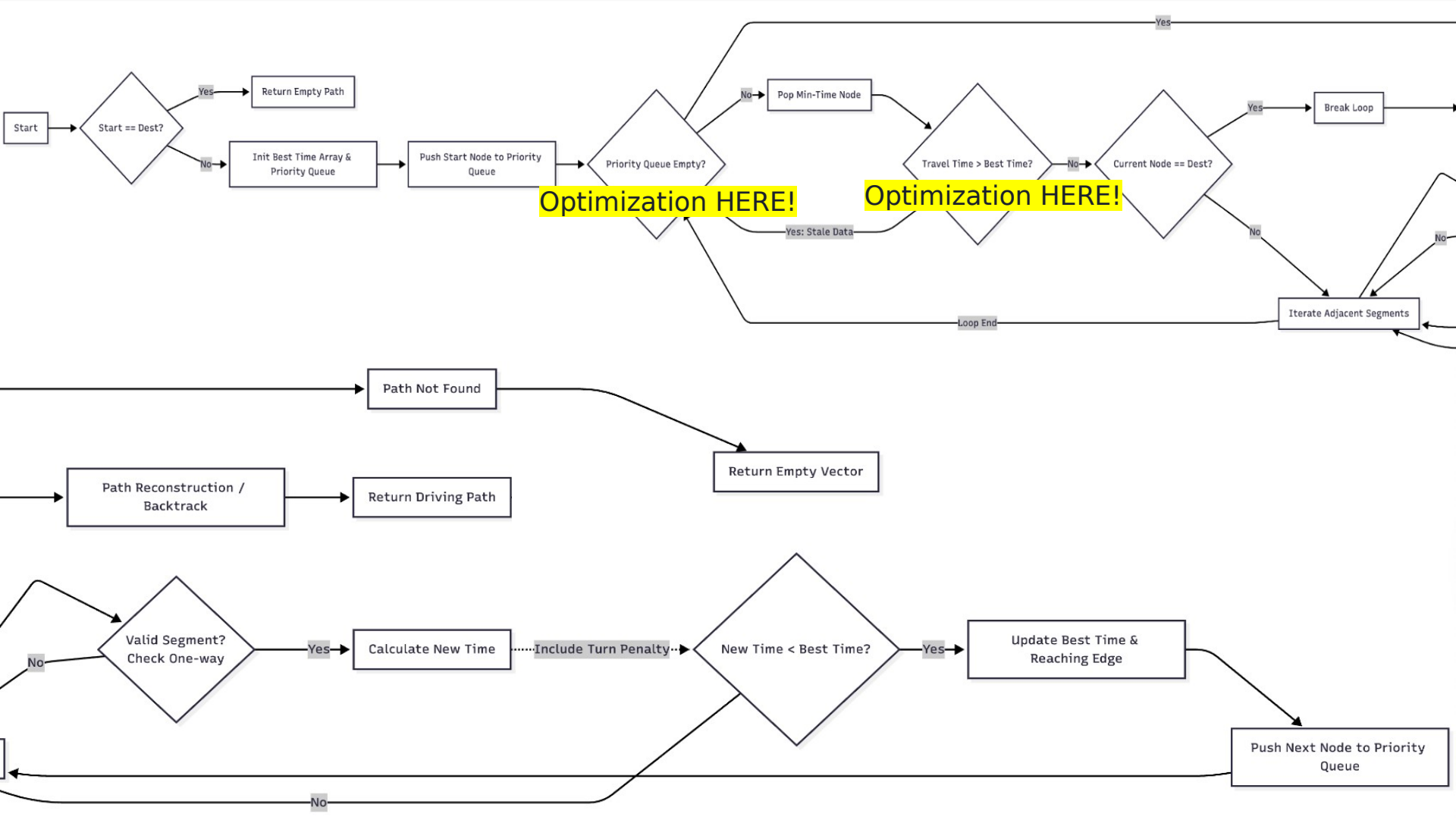
Trigonometry → Bad

Addition → Good

```
if (i > 0 && info.streetID != previous_street_id) total_travel_time += turn_penalty;
```

Dij (Priority Queue with Min Heap, and OPTIMIZATION)







Milestone 4

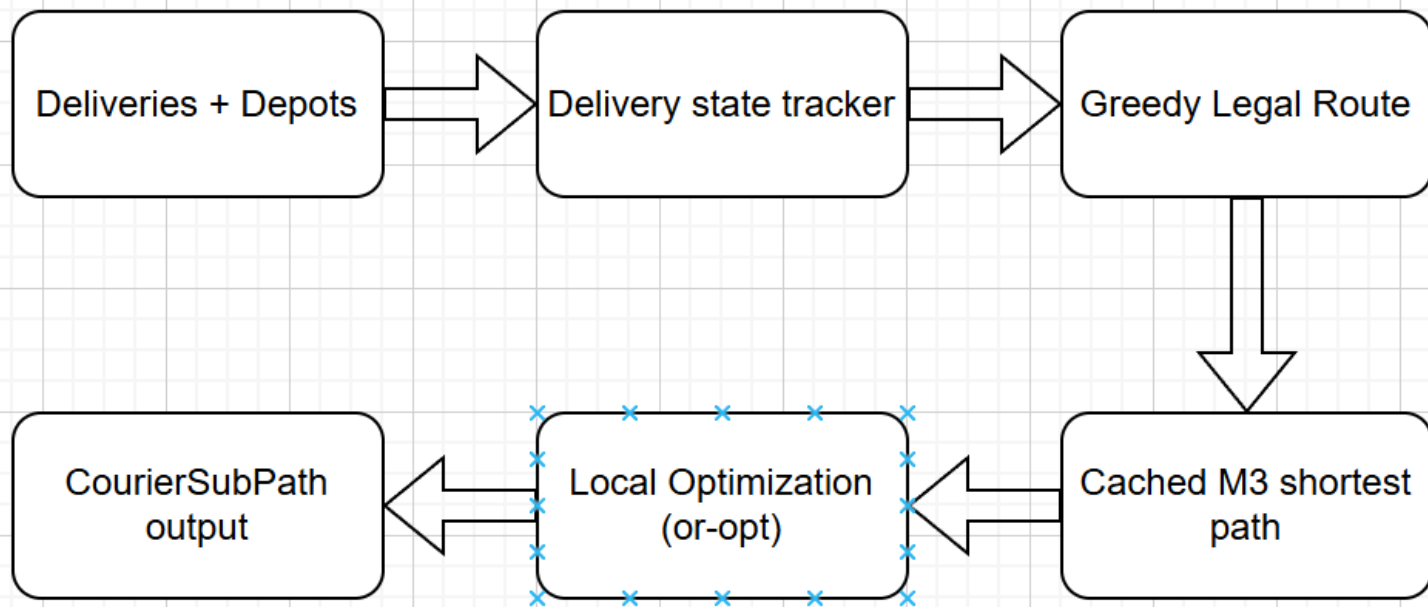
- M3 computes paths between intersections.
- M4 decides the order of pickup and drop-off visits.



Implementation: build first, then improve

- Track delivery state: Which packages are picked up, dropped off, or still pending
- Build an initial legal route: Greedy Algorithm
- Cache shortest paths: Reuse M3 pathfinding results
- Improve the route locally: Or-opt + parallel computation

Courier pipeline: from deliveries to route





Benchmark result: valid routes returned within the tester limit for 31 / 32 cases

Metric	Result
Total Benchmark Cases	32
Passed (within 50 sec for speed test)	31
Pass Rate	96.9%
Failed case	1 extreme-level case
Failed runtime	Approx. 51.4 sec
Time limit	50 sec



Future Improvements

- Avoid new Dijkstra calls near timeout
- Store partial path results from greedy phase for final conversion



III. Future Pitch

A Privacy First Personalized Map



Current maps hide how routes are chosen

- Most maps only use speed as the metric
- What about safety, exploration, and accessibility?
- Personalization often requires storing user data.
- Users should control both routing priorities and privacy level



Empowering Users to choose the route objective

- Standard Mode
- Safety Mode
- Exploration Mode
- Hybrid Mode



Safety Mode uses map-based risk proxies

Metrics for safety score:

- Lighting (night mode)
- Road type and speed limit
- Average Pedestrian Density
- Incident History
- Open business nearby
- Intersection Complexity



Exploration Mode ranks routes by POI value under a time budget

Metrics:

- Category relevance
- Distance from route
- User preference
- Popularity or rating
- Diversity



Accessibility filters

- Avoid stairs and inaccessible paths
- Prefer ramps, elevators, and curb cuts
- Highlight accessible entrances and facilities
- Can be combined with Standard, Safety, or Exploration Mode



Privacy controls decide what the map remembers

- Offline Mode
- Memory On
- Assistant On
- Personalization Off Switch



A map should adapt to the user, not the other way around

- Choose the route objective
- Control what data is remembered
- Understand why a route was recommended



IV. Conclusion



The Best Map Shows What Matters

Simplicity means:

- Less visual noise
- Less interaction friction
- Clearer route explanations
- More control over preferences and privacy



V. References



Reference:

- [1]City of Toronto, “Toronto Maps,” *map.toronto.ca*. <https://map.toronto.ca/torontomaps/>
- [2]“Baidu Map.” *Baidu.com*, 2013, map.baidu.com/.
- [3]Google, “Google Maps,” *Google Maps*, 2025. <https://www.google.com/maps/>
- [4] M. P. Peterson, *Online Maps with APIs and WebServices*. Springer Science & Business Media, 2012, pp. 157–175.
- [5]M.-C. Zheng and Y.-W. Hsu, “How 2.5D Maps Design Improve the Wayfinding Performance and Spatial Ability of Map Users,” *Informatics*, vol. 8, no. 4, p. 88, Dec. 2021, doi: <https://doi.org/10.3390/informatics8040088>.